



Potential biomarkers in hypoglycemic brain injury

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Abstract

Oxidative stress is a major underlying mechanism in hypoglycemic brain injury. Several oxidative stress-related proteins were identified through previous proteomics and literature review. The aim of the present study was to evaluate the potential of these proteins as biomarkers in hypoglycemic brain injury. Forty male Sprague Dawley rats were randomly and equally divided into four groups: control, acute hypoglycemia, hypoglycemia resuscitation 24 h, and hypoglycemia resuscitation 7 days. The hypoglycemic brain injury rat model was successfully constructed according to the Auer model. Real-time fluorescent quantitative polymerase chain reaction, western blot analysis, and immunohistochemical staining were used to quantify the expression of oxidative stress-related proteins. We also verified the expression level of selected protein in the brain samples of fatal insulin overdose cases. The expression of oxidative stress-related proteins PEX1/5/12 was down-regulated in hypoglycemic brain injury ($P < 0.05$), while the expressions of DJ-1 and NDRG1 were up-regulated ($P < 0.05$). Compared with the control group, the serum oxidative stress indexes SOD and MDA in the acute hypoglycemia group were significantly different ($P < 0.01$). The expressions of DJ-1 and NDRG1 in the hippocampus, cortex, and hypothalamus of rats were increased ($P < 0.05$). The expressions of DJ-1 and NDRG1 proteins in the cortex of the autopsy samples of insulin overdose were increased ($P < 0.05$). Oxidative stress-related proteins showed potential value as specific molecular markers in hypoglycemic brain injury, but further confirmatory studies are needed.

Keywords Oxidative stress · Hypoglycemic brain injury · Peroxisomes · DJ-1 · NDRG1

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Introduction

The discovery of insulin in 1921 was one of the most important biomedical events in the last century [1]. It is still the most effective drug for diabetes. Millions of people have insulin-dependent diabetes mellitus (DM), and the use of it far exceeds that of any other peptide or protein drug. It is approved as a treatment for almost all types of DM, and it is the last resort for patients with adult-onset DM [2].

Hypoglycemia is the leading complication in insulin therapy, and it is the main obstacle for patients with type 1 and type 2 DM to safely lower blood glucose levels [3]. Previous studies showed that severe hypoglycemia occurs less frequently during blood glucose control in subjects with type 2 DM compared with type 1 DM [4–6]. Mild to moderate and asymptomatic hypoglycemia affects most patients with well-controlled blood glucose levels [7]. On average, patients with type 1 DM experience 1 to 1.7 severe hypoglycemia events per year [8].

Recurrent severe hypoglycemia may cause brain injury and thus decrease cognitive function. The IQ scores of type 1 DM patients with a history of severe hypoglycemia were

significantly lower than control [9]. Magnetic resonance imaging (MRI) of type 1 DM patients with a history of more than 5 severe hypoglycemia events showed that half of them had cortical atrophy [10]. Computed tomography and MRI demonstrated that the basal ganglia, cerebral cortex, nigra, and hippocampus were the target regions affected by severe hypoglycemia [11]. Recent studies showed that both uncontrolled hyperglycemia and repeated hypoglycemia contributed to dementia in DM patients [12–14].

The wide availability of insulin as a drug and its colorless and tasteless properties make it a compound that can be used for homicide or suicide in forensic practice [15–18]. The cause of death in these cases is attributed to hypoglycemic brain injury. The hypoglycemic brain injury rat model was first developed by Auer and colleagues [19]. Deng et al. and Tong et al. conducted proteomic studies through the rat model, revealing the relevant mechanism of hypoglycemic brain damage and providing a reference for forensic diagnosis [20, 21]. Due to the critical role of oxidative stress in hypoglycemic brain injury, we selected oxidative stress-related proteins from their proteomic studies. Decreased expression of peroxisomal biogenesis factors (PEX)1/5/12 and increased expression of NDRG1 and DJ-1 were observed in their proteomic studies, and the verified of them were absent.

MDA is a product of lipid peroxidation that indicates lipid oxidation in the body. Serum MDA is a commonly used biomarker in oxidative stress [22–24]. SOD is the main antioxidant in the body and is associated with free radical metabolism. Decreased SOD levels can lead to the accumulation of free radicals and enhance oxidative stress. Increased serum Bax levels indicate the activation of apoptotic pathways under stress [25]. The MDA, SOD, and Bax levels in serum are closely associated with oxidative stress.

Peroxisomes are single-membrane-bound organelles that exist in almost all eukaryotic cells and participate in various metabolic pathways of catabolism and anabolism [26]. Peroxisomes play a role in the metabolism of oxygen, reactive oxygen species (ROS), and reactive nitrogen [27]. Peroxisomal biogenesis factors (PEX)1/5/12 are associated with protein transport, and dysfunction of these proteins may cause neuropathological abnormalities [28–30]. NDRG1 and DJ-1 are sensor proteins that respond to oxidative stress and protect cells from ROS and thus increase cell survival in cellular stress.

The present study of the hippocampus of rats with hypoglycemic brain injury revealed differential expression of peroxisomes and oxidative stress-related proteins, suggesting that oxidative stress-related proteins play an important role in hypoglycemic brain injury and may be potential molecular markers. The different expressed proteins were verified in animal models and autopsy samples to clarify the molecular mechanism of hypoglycemic brain injury that could help guide clinical treatment and forensic identification.

Materials and methods

Hypoglycemic brain injury rat model

Forty adult male Sprague Dawley rats (250–300 g) were provided by the Animal Center of Tongji Medical College, Huazhong University of Science and Technology. They were randomly divided into four equal groups: control, acute hypoglycemia, hypoglycemia resuscitation 24 h, and hypoglycemia resuscitation 7 days. The hypoglycemic brain injury rat model was successfully generated as described previously [19–21].

The rats were fasted overnight and injected with insulin (Novo 30R, Denmark) intraperitoneally at a dose of 15–20 IU/kg to induce hypoglycemia, and the blood glucose was detected every 30 min using the tail vein. After anesthesia with isoflurane gas (R510-22, RWD, China), endotracheal intubation was performed. Then, the cortical EEG was detected, and after continual equipotential EEG for 1 h, the rats were selectively resuscitated according to the experimental group. Only the hypoglycemia resuscitation 24 h group and 7 days group needed to be resuscitated and fed normally after resuscitation within the time of the experimental plan. The follow-up experiment was carried out directly in the acute hypoglycemia group. The rats in control were injected with the same dose of normal saline for subsequent experiments.

Resuscitation method is as follows: 50% glucose mixed with bicarbonate buffer 1:1 was injected intraperitoneally, and blood glucose was monitored continuously at the same time. After the rats were resuscitated, they were fed normally until the time specified in the experiment.

Autopsy samples from the Department of Forensic Medicine, Tongji Medical College, Huazhong University of Science and Technology were divided into a control group (coronary heart disease death) and an insulin overdose death group. The same frontal cortexes sized 2.0 cm × 2.0 cm × 4 mm were sampled and fixed in paraffin for the present study. The related information of these cases was summarized in Table 1.

Sample preparation

Rats were anesthetized, and the bottom of the eyeball was clamped with tweezers to quickly pull it out. A pre-prepared centrifuge tube was used to collect blood, which was centrifuged at 12,000 r/min at 4 °C, and the serum was placed in a refrigerator at – 80 °C until analysis.

Five rats in each group were perfused, and the brain was extracted and fixed in 4% paraformaldehyde. Paraffin sections were made and used for hematoxylin and eosin (HE) staining.

Table 1 Related information on autopsy cases

	Case No	Age/sex	Interval between insulin injected and death	Postmortem interval	Antemortem psychological status
Death due to insulin overdose	1	55/male	Within 8 h	3 days	Coma
	2	27/male	Within 10 h	10 days	Coma
	3	22/male	Within 30 h	2 days	Coma
	4	39/female	Within 8 h	7 days	Coma
Death due to coronary heart disease death	1	55/male		4 days	
	2	30/male		8 days	
	3	35/male		3 days	
	4	37/female		6 days	

Randomly perfuse 5 rats in each group, extract the brain, separate target parts of the brain, and then store at refrigerator at -80°C for inspection.

Enzyme-linked immunosorbent assays (ELISAs)

Determine the SOD, MDA, and Bax content in serum according to the instructions of the ELISA kit (AK061, Bioss; RA20003, RA20148, Bio-Swamp).

Fluorescent real-time quantitative polymerase chain reaction (RT-qPCR)

Four frozen hippocampi of each group were used to detect the relative mRNA levels of oxidative stress-related protein. The cDNA sequences of rat PEX1/5/12, DJ-1, and NDRG1 genes were downloaded from the NCBI database as reference design primer sequences, and GAPDH was used as an internal reference (Table 2). The designed specific primers were synthesized by Wuhan Servicebio. RNA was extracted from the hippocampus and diluted to 100–500 ng/ μl before conducting RT-qPCR. Two micrograms of RNA, 0.5 μl Oligo (dT)18 Primer, 0.5 μl random hexamer primer, and ddH₂O were added to make the volume 15 μl . It is then kept at 65°C for 5 min, quickly refrigerated, added with 4 μl 5 $\times$ Reaction Buffer and 1 μl Wuhan Seville ®RT Enzyme Mixa, kept at 42°C for 60 min, and then kept at 70°C for 5 min to inactivate reverse transcriptase. It is then added with 7.5 μl of 2 $\times$ qPCR Mix, 1.5 μl of 2.5 μM gene primers, 2.0 μl of reverse transcript product, and 4.0 μl of ddH₂O to configure the reaction system, and it is then first kept at 95°C for 10 min and 95°C for 15 s $\rightarrow$ 60°C for 40 cycles; set the melting curve from 60 to 0.3°C every 15 s, and finally, increase it to 95°C . Every sample is repeated three times.

Western blotting

Samples were prepared, and protein concentrations were quantified prior to western blotting. Sodium dodecyl sulfate (SDS) gels were prepared according to the gel kit instructions (20324ES62, 20326ES62, Yeasen), and protein samples were separated under 80 V constant voltage electrophoresis until bromophenol blue reached the junction of the concentrated and separation gels. The samples were then run at 120 V constant pressure electrophoresis, which was stopped as soon as they reached the bottom of the separation gel.

Target gel bands were isolated according to the molecular weight of the target protein and transferred to polyvinylidene fluoride (PVDF) membranes under a constant current of 250 mA for 150 min. The PVDF membranes were incubated in 5% skim milk for 2 h and then incubated in

Table 2 Primers used for qPCR

	Forward primer 5' $\rightarrow$ 3'	Amplified fragment length
<i>GAPDH-F</i>	CTGGAGAAACCTGCCAAGTATG	138 bp
<i>GAPDH-R</i>	GGTGGAAGAATGGGAGTTGCT	
<i>PEX1-F</i>	CTGTCCCAGTCTCCAGGTCATA	315 bp
<i>PEX1-R</i>	CACAGCTTTTACCTCTTCTTCACTC	
<i>PEX5-F</i>	TTGGCTAACTCTGAGGGCACA	212 bp
<i>PEX5-R</i>	GTGAGGTCGTCATAGTCAGAAAGC	
<i>PEX12-F</i>	TACAGAGTCTTCTGGCTGCCTAC	218 bp
<i>PEX12-R</i>	ATGCTTCTGACGGGCTCCTG	
<i>DJ-1-F</i>	GCCGTGATGTAGTGATTTGTCC	247 bp
<i>DJ-1-R</i>	GCCAATGGGTGCGATGTAA	
<i>NDRG1-F</i>	AGGGCAACCGTCCTGTCAT	213 bp
<i>NDRG1-R</i>	CAGGAAGCATTTCAGCCAAC	

diluted primary antibody DJ-1 (1:5000, ab76008, Abcam), NDRG1 (1:2000, DF6865, Affinity), PEX1 (1:1000, ab105027, Abcam), PEX5 (1:5000, ab192322, Abcam), PEX12 (1:2000, PA5-68,265, Invitrogen), and GAPDH (1:100,000, GB15004, Servicebio) at 4 °C overnight. The next day, PVDF membranes were incubated in horseradish peroxidase (HRP)-labeled donkey anti-goat immunoglobulin G (IgG) (1:10,000, GB23404, Servicebio) or HRP-labeled goat anti-rabbit IgG (1:8000, GB23303, Servicebio) at room temperature for 1 h. The strips were imaged in a chemiluminescence machine, and ImageJ software was used to measure the gray values of the images.

HE and immunohistochemistry staining

The slides were subjected to gradient dehydration, blocked with 3% hydrogen peroxide for 10–15 min to block endogenous enzymes, immersed in retrieval buffer (citrate buffer), and heated in a microwave oven with an interval of 5 to 10 min; this process was repeated 1 to 2 times. The tissue on the slide was sealed with an immunohistochemical pen, blocked in 5% bovine serum albumin, and kept at room temperature for 30 min. They were then incubated in primary rabbit monoclonal antibody against DJ-1 (1:1000, ab76008, Abcam), NDRG1 (1:200, DF6865, Affinity), PEX1 (1:200, 13,669–1-AP, Invitrogen), PEX5 (1:200, 12,545–1-AP, Invitrogen), and PEX12 (1:200, PA5-68265, Invitrogen) overnight at 4 °C. Next, biotinylated goat anti-rabbit IgG was added dropwise and incubated at 37 °C for 30 min. We then performed diaminobenzidine staining, re-stained with hematoxylin dye solution, and added hematoxylin color

separation solution. Following gradient dehydration and transparent xylene treatment, the slides were sealed with neutral resin.

For each section, three fields were randomly selected to collect $\times 200$ microscopic images. Image Pro Plus software was used to assess the average optical densities (ODs) of PEX1/5/12, DJ-1, and NDRG1.

Results

Serum ELISAs

The SOD level of the acute hypoglycemia group was lowest in all groups and then gradually increased with recovery time (Fig. 1A). The MDA level of the acute hypoglycemia group was higher than control and gradually decreased with recovery time (Fig. 1B). The serum Bax level was highest in the acute hypoglycemia group and then decreased slightly in the recovered 24 h group (Fig. 1C).

Fluorescence RT-qPCR

PEX1/5/12 mRNA levels decreased to different degrees in the acute hypoglycemia and recovered groups. PEX1 mRNA levels in the acute hypoglycemia group were significantly lower than control (Fig. 2A). PEX5/12 mRNA expression decreased significantly in the recovered 24 h group (Fig. 2B, C). DJ-1 and NDRG1 mRNA expression was significantly increased in the acute hypoglycemia group but decreased in the recovered group (Fig. 2D, E).

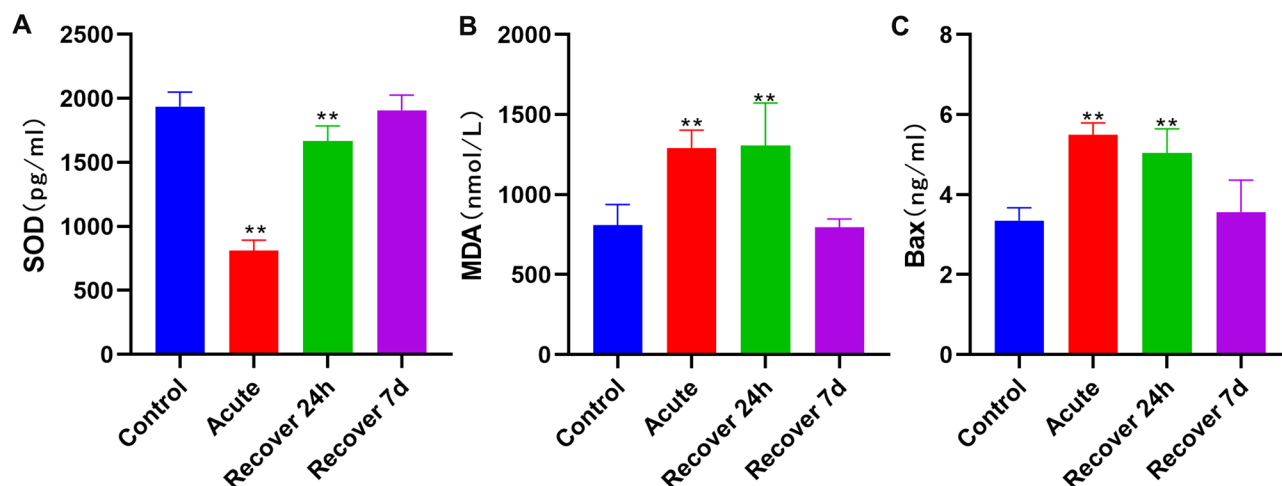


Fig. 1 Changes in serum levels of **A** MDA, **B** SOD, and **C** Bax measured by ELISA. One asterisk denotes $P < 0.05$; two asterisks denote $P < 0.01$

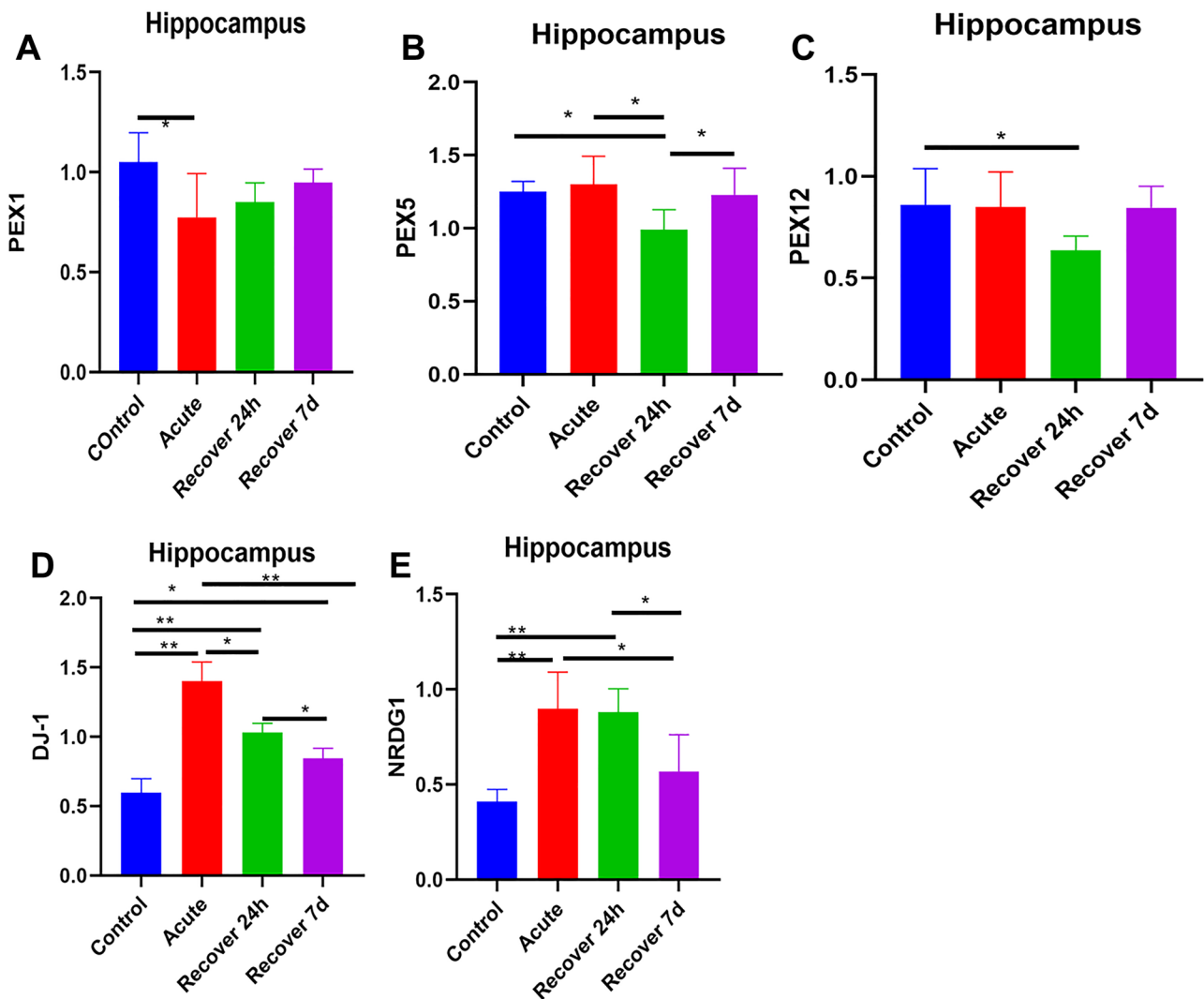


Fig. 2 mRNA levels of **A** PEX1, **B** PEX5, **C** PEX12, **D** DJ-1, and **E** NDRG1 in the rat hippocampus. One asterisk denotes $P < 0.05$; two asterisks denote $P < 0.01$

Western blot

Hippocampus

The expression of PEX1/5/12 in the hippocampus decreased in the acute hypoglycemia and recovered 24 h groups and increased in the recovered 7 days group (Fig. 3). The expression of PEX1/12 was reduced in the recovered 24 h group, and the expression of PEX5 was reduced in the acute hypoglycemia group (Fig. 3A–D). The expression of DJ-1 and NDRG1 in the hippocampus was increased in the acute hypoglycemia and recovered 24 h groups (Fig. 3E–G).

The expression of DJ-1 and NDRG1 in the cortex and hypothalamus was significantly increased in the acute hypoglycemia group and gradually decreased in the recovered groups (Fig. 4).

HE and immunohistochemistry staining

Severe cerebral edema and cortical neuron necrosis were seen in the acute hypoglycemia and recovered 24 h groups. Hippocampal neuron necrosis was observed in the acute hypoglycemia, recovered 24 h, and 7 days groups (Fig. 5). DJ-1 was mainly expressed in granule neurons and glial cells, and NDRG1 was mainly expressed in glial cells (Fig. 6). The average ODs of DJ-1 and NDRG1 in the hippocampus and cortex of the acute hypoglycemic and recovered 24 h groups were higher than control (Fig. 7).

Validation in human autopsy cases

DJ-1 was mainly expressed in neurons and glial cells, and NDRG1 was mainly expressed in glial cells (Fig. 8). The

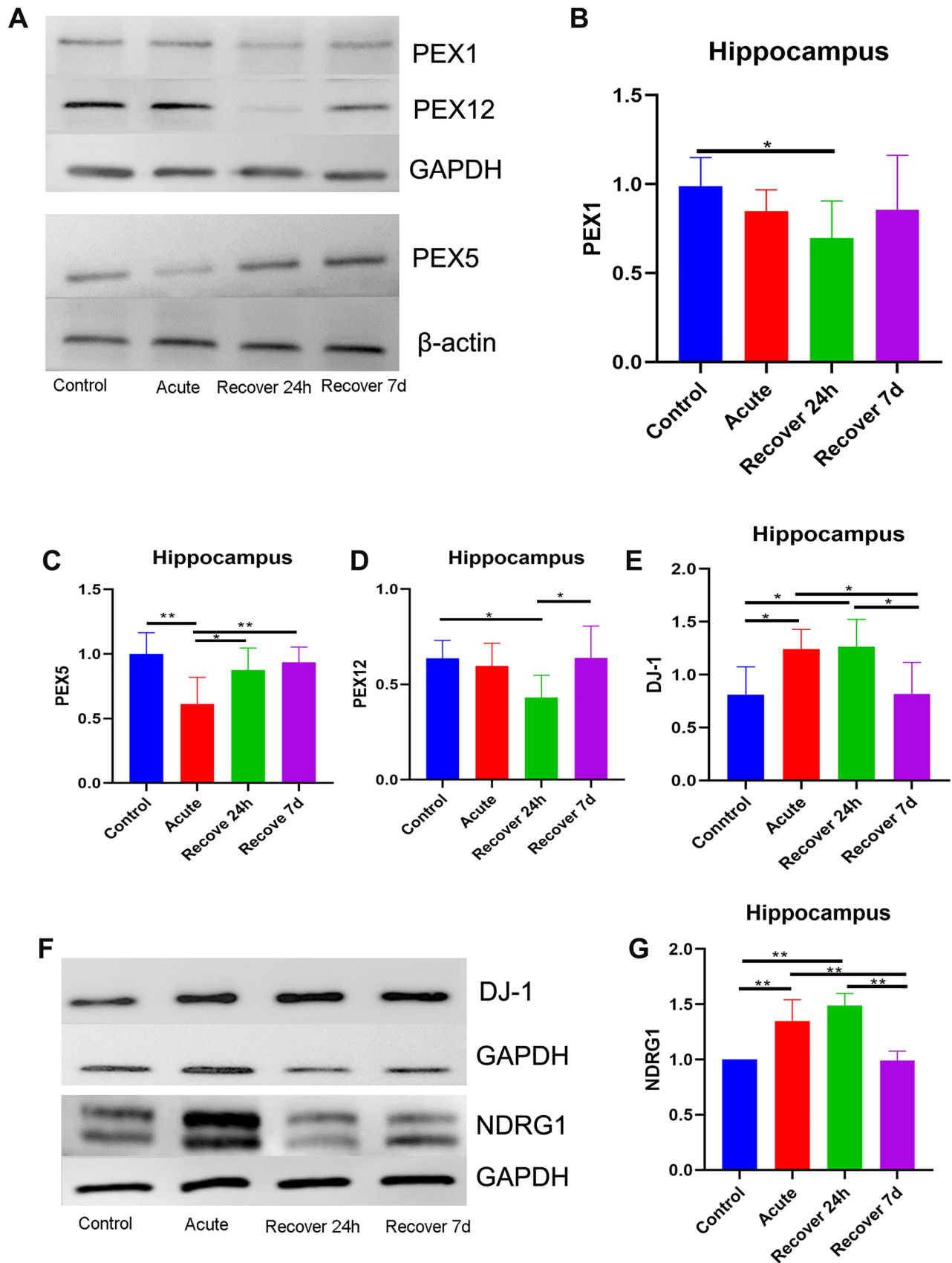


Fig. 3 Western blot results of PEX1/5/12, DJ-1, and NDRG1 in the rat hippocampus. **A** Western blot image of PEX1/5/12, DJ-1, and NDRG1. Quantification of **B** PEX1, **C** PEX5, and **D** PEX12. **E** Western blot image of DJ-1. Quantification of **F** PEX12 and **G** NDRG1. One asterisk denotes $P < 0.05$; two asterisks denote $P < 0.01$

average ODs of DJ-1 and NDRG1 in the hippocampus and cortex of the acute hypoglycemic and recovered 24 h groups were higher than control (Fig. 8).

Discussion

Hypoglycemic brain injury, which was common in diabetic patients with insulin therapy, is rare in forensic practice. The hypoglycemic brain injury in forensic practice was often associated with insulin overdose suicide or homicide [15–18, 31]. Birkinshaw reported the first homicide case of an insulin overdose in 1958, and since then, the insulin overdose-related death came to the attention of forensic pathologists [32]. Although many studies focused on hypoglycemic brain injury from a forensic perspective, it is still a challenge in forensic practice, and no reliable biomarkers were confirmed [32].

MDA and SOD were the common biomarkers in oxidative stress. Increased serum MDA and decreased SOD levels indicate the existence of oxidative stress during hypoglycemic brain injury, and increased serum Bax suggests the activation of apoptotic pathways during hypoglycemic brain injury, which was consistent with the pathological changes in hypoglycemic brain injury and indicated the critical role of oxidative stress in hypoglycemic brain injury.

To find the biomarkers of hypoglycemic brain injury, Deng et al. conducted proteomic studies on the cortical tissue of hypoglycemic brain injury rats to screen different expressed proteins, and NDRG1, DJ-1, AQP4, Inpp4a, and Clusterin were verified through RT-PCR and western blot and may be used as molecular markers in hypoglycemic brain injury [20]. Further studies have shown that DJ-1 can protect neurons from hypoglycemia by regulating autophagy, and AQP4 can increase brain edema by regulating inflammatory responses, whereas these molecular markers were not validated in autopsy samples [33].

NDRG1 is a poly-phosphorylated protein, primarily located in the cytoplasm [34]. Its expression is regulated by a variety of endogenous and environmental signals and is up-regulated to increase cell survival in cellular stress [35, 36]. NDRG1 is an important protein in myelinating cells and plays an important role in myelination formation; its dysregulation is highly associated with neurological diseases [37]. In clinical studies, NDRG1 protein is a biomarker for cancer diagnosis and is widely used in various cancer diagnoses [38]. The mutations in NDRG1 are associated with

subcortical white matter abnormalities and severe demyelinating neuropathy [39]. The increased expression of NDRG1 protein in patients with cortical dysplasia was attributed to oligodendrocyte proliferation [40]. Approximately 14% of cortical dysplasia patients suffer from epilepsy, and abnormally elevated NDRG1 was identified as the underlying mechanism [41]. The up-regulation of NDRG1 expression during hypoglycemia may be related to the proliferation of gliocytes, promote cell survival, and may play a role in epilepsy during extreme hypoglycemia, so it may serve as a potential molecular marker in hypoglycemic brain injury.

DJ-1 is widely expressed in neurons and glial cells; it is mainly localized in the cytoplasm but trans-locates to the mitochondria to play a cytoprotective role under oxidative stress [42–45]. DJ-1 has been used as a biomarker for oxidative stress-related diseases including Parkinson's disease, and elevated DJ-1 in the saliva is associated with Parkinson's disease [46]. During hypoglycemic brain injury, the overexpression of DJ-1 protein indicates that there is severe oxidative stress in the hippocampus. Previous studies have shown that exogenous DJ-1 protein can reduce oxidative stress during brain injury and thus play a neuroprotective role, suggesting that DJ-1 protein has a potential protective role in hypoglycemic brain injury and can be used as a potential molecule for hypoglycemic brain injury.

Peroxisomes are crucial in the production and removal of H_2O_2 , and they trigger oxidative stress and ROS signaling [47]. Peroxisomal proteins are synthesized in the cytoplasmic ribosome and rapidly transported to organelles through shuttle receptors [48]. PEX5 plays a critical role in protein transport and is triggered by the cellular redox state [49]. On one hand, unbalanced protein transport in peroxisomes in the setting of oxidative stress affects ROS clearance, thus aggravating oxidative stress and resulting in peroxisome dysfunction; on the other hand, peroxisomes also affect the integrity of the inner mitochondrial membrane, and peroxisome dysfunction can reduce the activities of complexes I, III, and IV, leading to more pronounced oxidative stress damage [50–52]. The changes of PEX1/5/12 verified through multiple methods were consistent with Tong's proteomics study and were associated with oxidative stress damage.

Paradoxically, PEX1/5 deficiency can protect cells from oxidative stress damage [53, 54]. Various clinical cohort studies demonstrated that peroxisomes play a crucial role in neurodegenerative conditions such as Alzheimer's disease [55–57]. The reduced PEX1/5/12 level in hypoglycemic brain injury may initially aggravate oxidative stress injury and protect the brain from oxidative stress injury at last, which needs further study to confirm.

Peroxisomes can participate in the biosynthesis of ether phospholipid and docosahexaenoic acid and thus play important roles in signal transduction and nervous system development [58, 59]. Peroxisome biogenesis disorders are

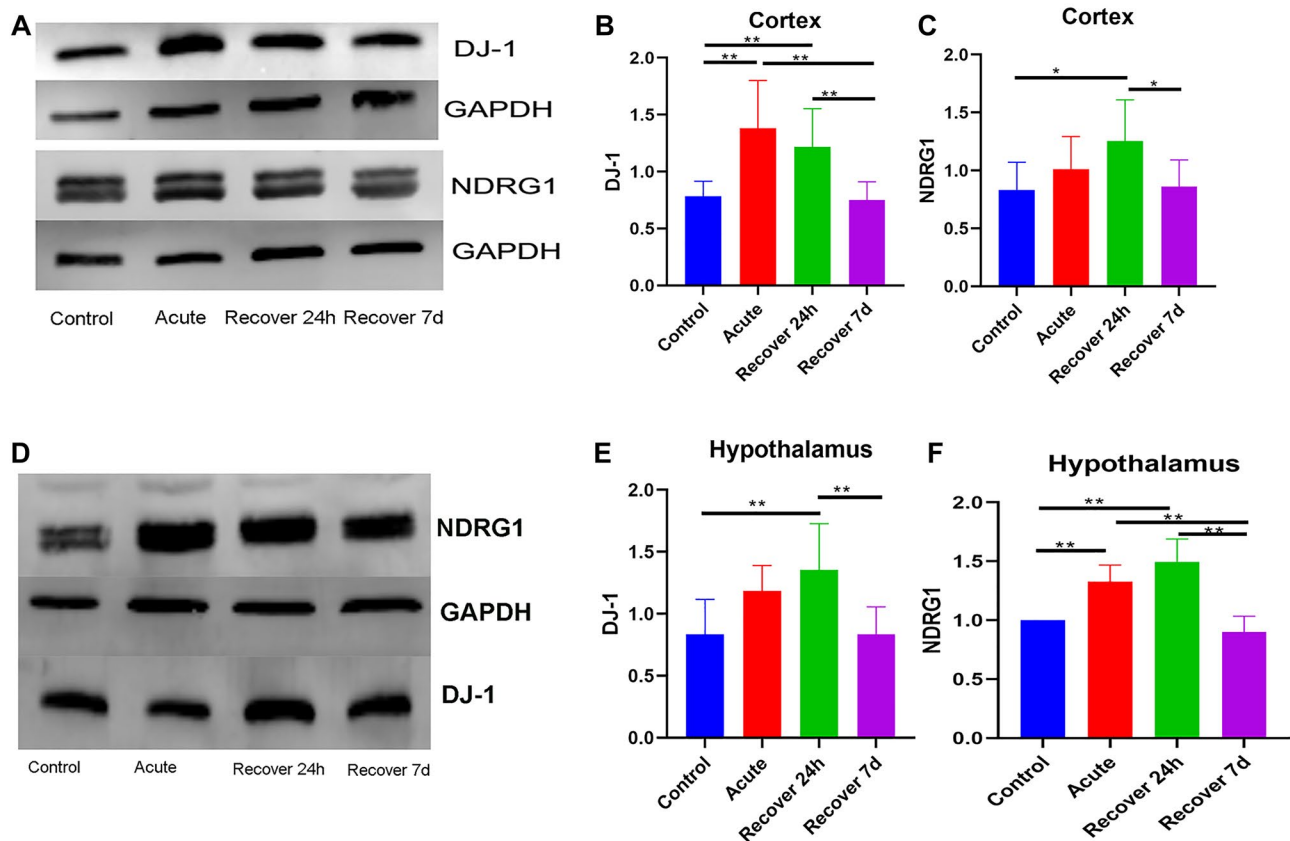


Fig. 4 Western blot results of DJ-1 and NDRG1 in the cortex and hypothalamus of rat models. **A** Blots of DJ-1 and NDRG1 in the cortex; quantification of **B** DJ-1 and **C** NDRG1. **D** Blots of DJ-1

and NDRG1 in the hypothalamus; quantification of **E** DJ-1 and **F** NDRG1. One asterisk denotes $P < 0.05$; two asterisks denote $P < 0.01$

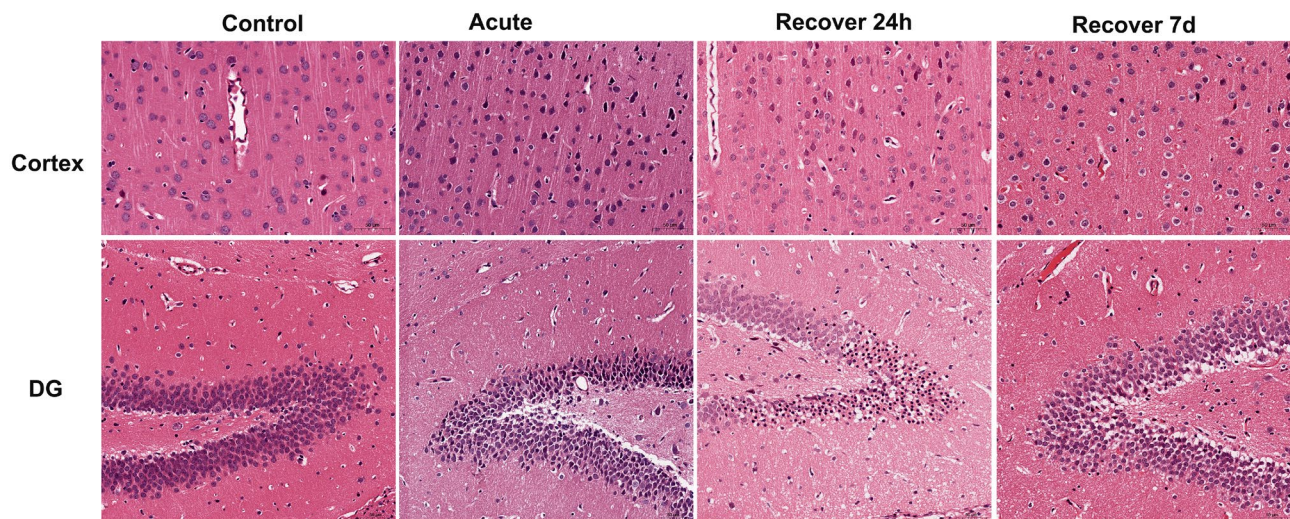


Fig. 5 The pathological changes in hypoglycemic brain injury which revealed severe cerebral edema and cortical neuron necrosis in the cortex, neuron necrosis in the hippocampus. HE staining $\times 200$

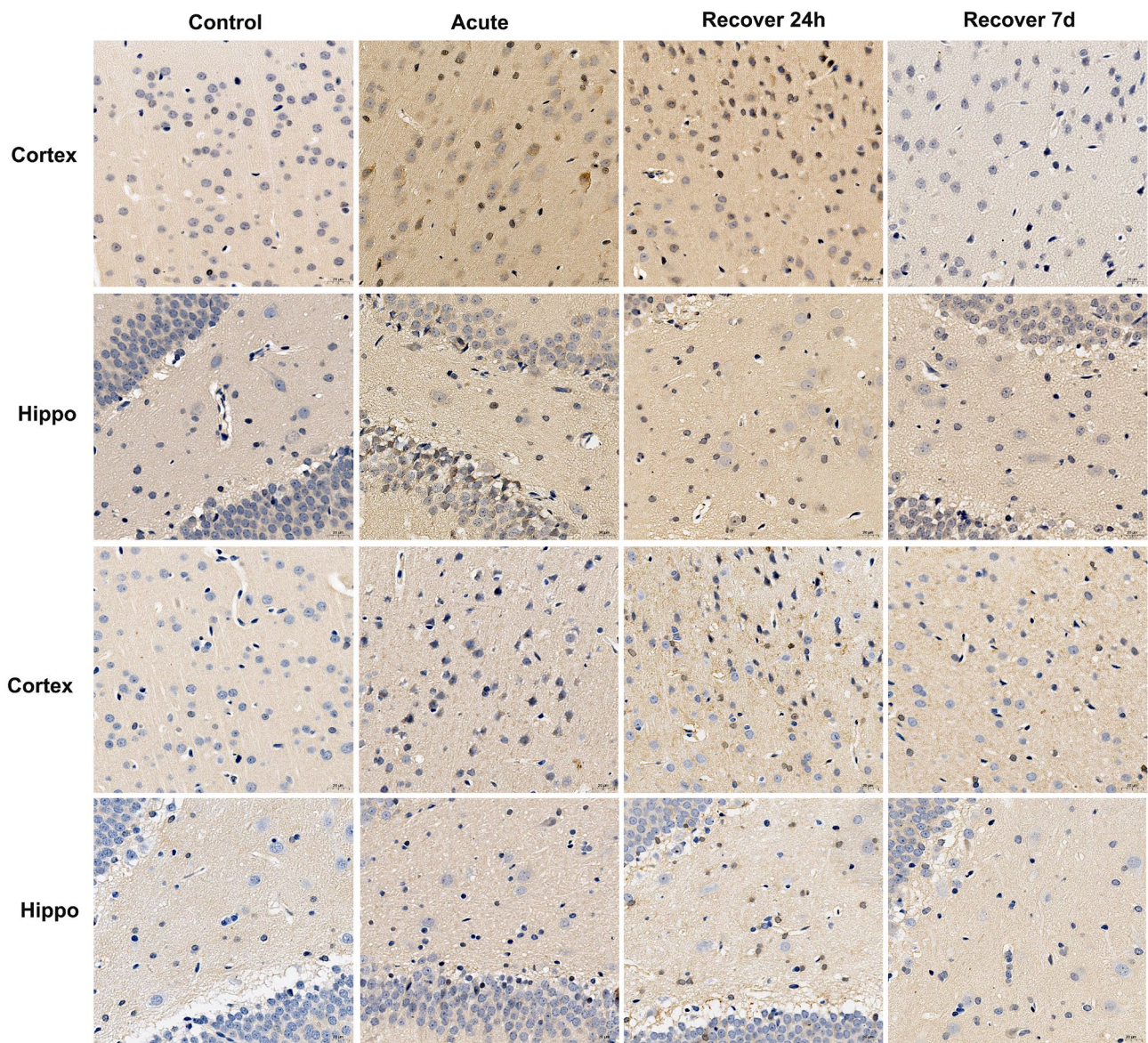


Fig. 6 The IHC results of DJ-1 and NDRG1 in rat models. DJ-1 was mainly expressed in granule neurons and glial cells in the cortex and hippocampus (the top half of the picture), and NDRG1 was mainly

expressed in glial cells in the cortex and hippocampus (the down half of the picture). DJ-1 staining and NDRG1 staining, $\times 200$

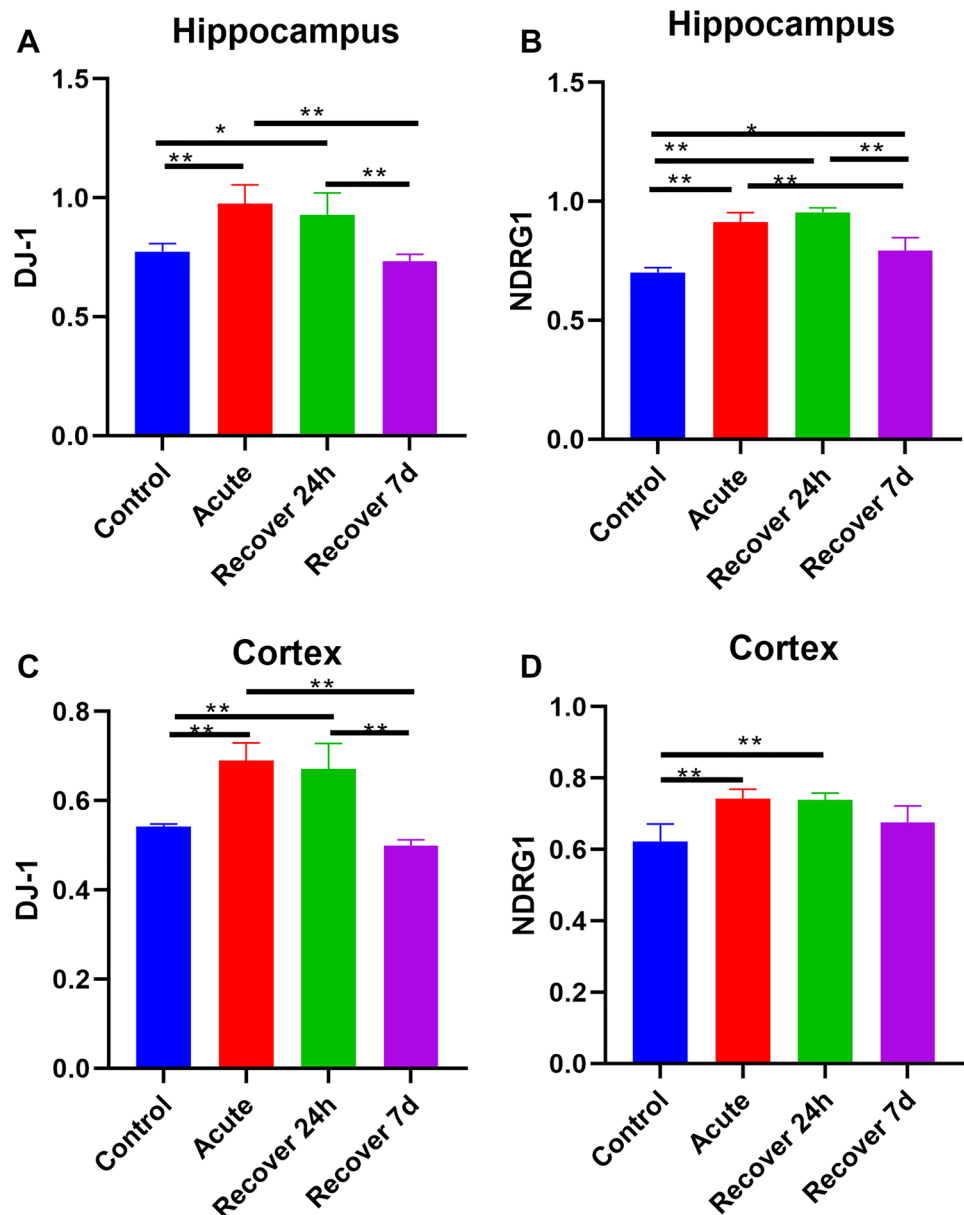
serious human diseases caused by reduced or absent peroxisome function, and PEX1 mutations are the most common cause [27, 60, 61]. Brain pathology in peroxisome diseases can be divided into three main categories: (i) abnormal neuronal migration or differentiation, (ii) defects in the formation or maintenance of central white matter, and (iii) post-developmental neuronal degeneration [62].

Previous study revealed swollen, disorganized, disrupted axons and astrocyte proliferation in hypoglycemia rat models and insulin overdose autopsy cases, and the pathological changes of axons may be associated with decreased expression of PEX1/5/12 and increased expression of NDRG1 [15, 63]. The increased NDRG1 and DJ-1 in astrocyte

proliferation may affect astrocyte-mediated neuroprotection by reducing ROS or inducing astrocytes to synthesize and release protective molecules [64–66].

Although NDRG1 and DJ-1 were confirmed in previous studies, the IHC confirmation of them and the verified autopsy samples were absent, present research further validated NDRG1 and DJ-1 expressions on animal models and autopsy specimens, indicating them as biomarkers in hypoglycemic brain injury. The validation of PEX1/5/12 was also conducted in animal models, whereas the performance of PEX1/5/12 antibodies in IHC limited the validation of them in human autopsy samples, which needs further study to confirm.

Fig. 7 The validation of DJ-1 and NDRG1 in hypoglycemic brain injury rats displayed the increased ODs of DJ-1 and NDRG1 in the hippocampus and cortex of the acute hypoglycemic and recovered 24h groups

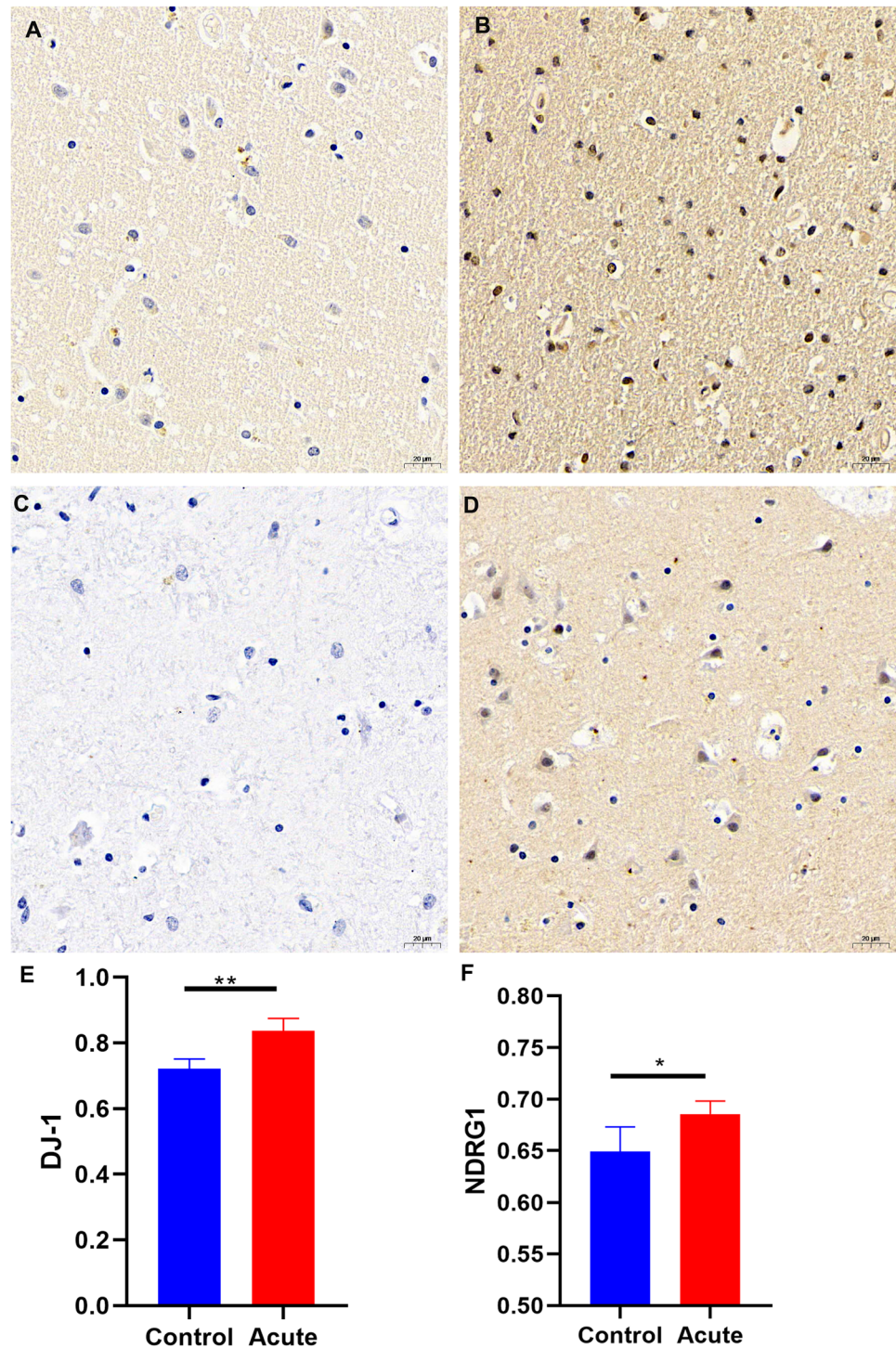


Limitation

Although the present study verified the potential role of PEX1/5/12, NDRG1, and DJ-1 as biomarkers in hypoglycemic brain injury through rat models and autopsy samples, limitations also existed in the present study. Firstly, cross-validation was absent, and the changes of them in serum and other brain regions were also absent; the validation of these results was needed to identify their role

as selected biomarkers. Secondly, the selected biomarkers and the involved human autopsy cases were small in the present study, and other potential biomarkers and more autopsy cases should be conducted. Finally, the age, sex rate, and postmortem interval of human autopsy cases involved in the present study may play a role in the validation of the selected biomarkers, but we conducted the same select condition on the control group to minimize their effect.

Fig. 8 The validation of DJ-1 and NDRG1 in insulin overdose death cases. **A, B**, DJ-1 was mainly expressed in neurons and glial cells. **C, D**, NDRG1 was mainly expressed in glial cells. **E**, The average ODs of DJ-1 in the cortex of the insulin overdose death cases were higher. **F**, The average ODs of NDRG1 in the cortex of the insulin overdose death cases were higher. **A, C**, the control group. **B, D**, the insulin overdose death group. **A, B**, DJ-1 staining, **C, D**, NDRG1 staining, $\times 200$



Conclusion

The successful validation of PEX1/5/12, NDRG1, and DJ-1 proteins in animal and insulin overdose brain specimens suggests that they can be used as biomarkers in forensic and clinical diagnosis. However, further studies with more specimens are still needed to verify their practical application.

Key points:

1. Oxidative stress is a major mechanism in hypoglycemic brain injury.
2. PEX1/5/12, DJ-1, and NDRG1 showed the same trend as proteomics research and literature in rat models and autopsy samples.

3. PEX1/5/12, DJ-1, and NDRG1 may be potential biomarkers in insulin overdose cases.

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Data availability The authors confirm that the data supporting the findings of this study are available within the article.

Declarations

Ethical approval All procedures performed in this study involving human participants and animals were approved by the Ethics Committee of Tongji Medical College, HUST.

Conflict of interest The authors declare no competing interests.

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